

7(a)	energy transformed from <u>chemical to electrical</u> / unit charge (driven around a complete circuit)	B1
7(b)(i)	the current decreases (as resistance of Y increases)	M1
	lost volts go down (as resistance of Y increases)	M1
	p.d. AB increases (as resistance of Y increases)	A1
7(b)(ii)1.	$1.50 = 0.180 \times (6.00 + 0.200 + R_X)$	C1
	$R_X = 2.1(3) \Omega$	A1
7(b)(ii)2.	p.d. AB = $1.5 - (0.180 \times 0.200)$ or $0.18 \times (2.13 + 6.00)$	C1
	= 1.46(4) V	A1
7(b)(ii)3.	efficiency = (useful) power output / (total) power input or IV / IE	C1
	(= $1.46 / 1.5$) = 0.97 [0.98 if full figures used]	A1

Question	Answer	Marks
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